

Table 1: Method comparison on CORE4D-G1 (all 50 paper sequences), mean over cases. **SPIDER-CEM** (ours) vs **SBTO** and **GMR**. Physical contact/penetration are frame fractions from the same MuJoCo contact detector; max penetration is the deepest hand-object interpenetration. $\uparrow/\downarrow$: higher/lower is better; best per row in bold. Note GMR’s low penetration@5mm is an artifact of barely contacting the object (contact@5mm $\approx$ 0.24%), not clean contact: its max penetration (23.47 mm) still exceeds ours (12.01 mm).

Metric	SPIDER-CEM	SBTO	GMR
Phys. contact@5mm (%) $\uparrow$	72.47	34.37	0.22
Phys. penetration@5mm (%) $\downarrow$	7.59	36.59	4.65
Max phys penetration (mm) $\downarrow$	12.01	25.36	23.51
Obj speed max (m/s) $\downarrow$	1.29	1.76	1.83
Foot slip max (m) $\downarrow$	0.87	1.20	0.97